

CS471 Project 4

Flow Shop Scheduling Using the NEH Heuristic and Metaheuristics

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1 Introduction

Scheduling problems are fundamental optimization problems that arise in manufacturing systems, production planning, and computer systems. One well-known scheduling problem is the Flow Shop Scheduling Problem (FSS), where a set of jobs must be processed on multiple machines in the same order. The objective of the problem is to minimize the makespan, which is the completion time of the final job on the final machine.

In this project, the Flow Shop Scheduling Problem was solved using the NEH heuristic across the 120 benchmark instances introduced by Taillard. These benchmark problems vary in size from small instances containing 20 jobs to large instances containing 500 jobs.

In addition to the standard flow shop scheduling problem, this project also considers the Flow Shop Scheduling Problem with Blocking (FSSB). In this variant, jobs cannot leave a machine until the next machine becomes available, which introduces additional constraints and often increases the makespan.

The NEH heuristic was implemented to construct schedules for both FSS and FSSB. A meta-heuristic based on Ant Colony Optimization (ACO) was then applied in an attempt to further improve the schedules generated by NEH.

2 Flow Shop Scheduling Problems

2.1 Standard Flow Shop Scheduling (FSS)

In the standard flow shop scheduling problem, each job must be processed on m machines in the same order. Let p_{ij} represent the processing time of job j on machine i .

A schedule is represented by a permutation of jobs:

$$\pi = (\pi(1), \pi(2), \dots, \pi(n))$$

The objective is to minimize the makespan:

$$C_{max}(\pi)$$

which represents the completion time of the last job in the sequence on the final machine.

The makespan for a given permutation can be computed efficiently in $O(mn)$ time.

2.2 Flow Shop Scheduling with Blocking (FSSB)

In the blocking flow shop scheduling problem, buffers between machines are not allowed. A job that finishes processing on a machine cannot leave that machine until the next machine becomes available.

This causes machines to become blocked when downstream machines are occupied. As a result, blocking constraints typically increase the makespan compared to the standard flow shop problem.

3 The NEH Heuristic

The NEH heuristic, proposed by Nawaz, Ensore, and Ham, is one of the most effective constructive heuristics for solving permutation flow shop scheduling problems.

The NEH algorithm operates as follows:

1. Compute the total processing time of each job:

$$T_k = \sum_{i=1}^m p_{ik}$$

2. Sort the jobs in decreasing order of T_k .
3. Take the first two jobs and determine the order that produces the smallest makespan.
4. Iteratively insert each remaining job into every possible position of the partial schedule.
5. Select the insertion that produces the smallest makespan.

This process continues until all jobs have been inserted into the schedule.

The NEH heuristic is known for producing high-quality solutions with extremely low computational cost.

4 Metaheuristic: Ant Colony Optimization

To further improve the schedules generated by the NEH heuristic, an Ant Colony Optimization (ACO) metaheuristic was implemented, augmented with a local search procedure.

ACO is a population-based search algorithm inspired by the behavior of ants searching for food. Artificial ants construct candidate schedules by probabilistically selecting jobs based on pheromone trails and heuristic information, balancing exploration of new solutions with exploitation of previously successful ones.

In this project, the NEH schedule was used to seed the ACO algorithm, providing a high-quality initial solution. Each ant then explored variations of the job permutation by constructing new schedules influenced by pheromone intensities and heuristic desirability.

To enhance solution quality, a local search procedure was applied to the schedules generated by the ants. This local search refined candidate solutions by performing neighborhood-based improvements (e.g., job swaps or insertions), allowing the algorithm to further reduce the makespan beyond what ACO alone could achieve. The combination of ACO's global search capability and local search's fine-tuning ability resulted in a more effective hybrid optimization approach.

Pheromone values were updated iteratively based on the quality of the solutions found, reinforcing better schedules over time. The objective of the ACO algorithm, combined with local search, was to refine the NEH schedules and achieve lower makespan values across the problem instances.

5 Experimental Results

The algorithms were tested on the 120 Taillard benchmark instances. These instances range from 20 jobs to 500 jobs and include different numbers of machines.

For each instance, the NEH heuristic generated an initial schedule. The ACO metaheuristic was then applied to attempt to improve this solution.

5.1 Complete Results for All 120 Taillard Instances

The following tables report the complete experimental results for all 120 Taillard benchmark instances. The results include the makespan and execution time obtained using the NEH heuristic and the Ant Colony Optimization (ACO) metaheuristic.

The data is automatically loaded from the experiment CSV output files generated by the program. This ensures that the tables reflect the exact computational results without requiring manual transcription.

5.2 Experimental Results

The following tables report the complete results for all 120 Taillard benchmark instances. The tables include the makespan and execution time for both the standard flow shop scheduling problem (FSS) and the blocking flow shop scheduling problem (FSSB).

The results are automatically generated from the experiment CSV output files produced by the program.

5.3 NEH Results

5.3.1 20 Job Instances (1–30)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
1	20	5	1319	0.116	1433	0.110
2	20	5	1365	0.076	1461	0.086
3	20	5	1165	0.075	1353	0.083
4	20	5	1329	0.076	1526	0.084
5	20	5	1305	0.077	1388	0.086
6	20	5	1269	0.076	1458	0.085
7	20	5	1256	0.109	1456	0.087
8	20	5	1227	0.080	1456	0.090
9	20	5	1284	0.078	1457	0.088
10	20	5	1153	0.076	1349	0.085
11	20	10	1668	0.130	1798	0.145
12	20	10	1764	0.153	1903	0.134
13	20	10	1558	0.131	1765	0.144
14	20	10	1418	0.127	1625	0.141
15	20	10	1502	0.130	1722	0.144
16	20	10	1453	0.115	1668	0.134
17	20	10	1542	0.128	1734	0.141
18	20	10	1620	0.135	1814	0.148
19	20	10	1664	0.128	1832	0.142
20	20	10	1656	0.133	1854	0.148
21	20	20	2404	0.234	2530	0.260
22	20	20	2127	0.243	2285	0.285
23	20	20	2401	0.231	2566	0.266
24	20	20	2262	0.256	2399	0.255
25	20	20	2397	0.224	2538	0.249
26	20	20	2349	0.226	2467	0.261
27	20	20	2362	0.236	2475	0.301
28	20	20	2249	0.220	2411	0.256
29	20	20	2306	0.220	2421	0.255
30	20	20	2277	0.235	2409	0.243

5.3.2 50 Job Instances (31–60)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
31	50	5	2730	1.112	3310	0.988
32	50	5	2882	1.290	3551	1.018
33	50	5	2625	0.909	3310	1.176
34	50	5	2783	0.992	3405	0.994
35	50	5	2876	0.930	3497	1.206
36	50	5	2842	0.884	3381	1.148
37	50	5	2802	1.083	3259	0.993
38	50	5	2688	1.220	3301	1.223
39	50	5	2575	0.917	3117	1.170
40	50	5	2786	0.909	3407	1.152
41	50	10	3133	1.875	3895	2.225
42	50	10	3042	1.707	3776	1.995
43	50	10	3029	1.607	3819	1.898
44	50	10	3158	1.746	3981	2.047
45	50	10	3154	1.615	3926	1.898
46	50	10	3127	1.586	3844	1.902
47	50	10	3311	1.587	3916	1.902
48	50	10	3140	1.706	3807	1.997
49	50	10	3017	1.528	3851	2.124
50	50	10	3258	1.629	3906	1.900
51	50	20	4054	3.053	4846	3.660
52	50	20	3909	3.158	4635	3.820
53	50	20	3913	3.360	4669	3.573
54	50	20	3956	3.130	4724	3.566
55	50	20	3906	3.314	4475	3.684
56	50	20	3888	2.967	4566	3.846
57	50	20	3968	3.012	4600	3.554
58	50	20	3966	3.364	4756	3.831
59	50	20	3918	3.155	4576	3.659
60	50	20	4015	3.262	4793	3.583

5.3.3 100 Job Instances (61–90)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
61	100	5	5535	7.208	6762	8.036
62	100	5	5288	6.731	6568	8.221
63	100	5	5219	7.119	6431	7.796
64	100	5	5023	6.645	6172	8.802
65	100	5	5266	6.691	6442	8.012
66	100	5	5139	7.131	6323	7.815
67	100	5	5272	7.053	6578	8.085
68	100	5	5129	6.908	6331	7.809
69	100	5	5489	6.957	6693	7.803
70	100	5	5334	6.595	6586	8.452
71	100	10	5947	12.601	7558	15.447
72	100	10	5402	12.467	7339	15.118
73	100	10	5740	12.016	7434	15.348
74	100	10	5904	11.764	7682	14.920
75	100	10	5599	11.627	7343	16.096
76	100	10	5409	11.700	7121	15.507
77	100	10	5745	11.596	7363	15.377
78	100	10	5712	13.730	7316	16.844
79	100	10	5998	12.108	7525	15.298
80	100	10	5923	11.603	7562	15.383
81	100	20	6625	24.055	8389	29.686
82	100	20	6592	24.870	8329	31.025
83	100	20	6634	24.254	8288	29.626
84	100	20	6621	23.621	8333	30.146
85	100	20	6685	23.421	8251	29.718
86	100	20	6722	24.049	8360	31.160
87	100	20	6580	23.979	8542	30.143
88	100	20	6800	24.944	8555	29.023
89	100	20	6662	23.525	8303	29.624
90	100	20	6767	24.039	8472	30.554

5.3.4 200 Job Instances (91–110)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
91	200	10	10976	94.365	14480	120.149
92	200	10	10639	98.657	14433	126.565
93	200	10	11072	96.169	14338	121.296
94	200	10	11057	93.807	14429	120.321
95	200	10	10613	101.141	14426	125.072
96	200	10	10449	94.742	14172	122.446
97	200	10	10960	101.294	14604	123.577
98	200	10	10871	97.672	14562	120.800
99	200	10	10558	98.169	14316	123.505
100	200	10	10794	100.891	14412	123.210
101	200	20	11620	203.491	15768	244.630
102	200	20	11749	204.203	15759	235.875
103	200	20	11839	203.526	15993	239.725
104	200	20	11816	206.825	15932	235.895
105	200	20	11628	212.907	15855	254.848
106	200	20	11717	216.937	15911	248.512
107	200	20	11776	201.220	15900	235.339
108	200	20	11855	202.658	15875	235.715
109	200	20	11659	204.404	15997	241.504
110	200	20	11893	200.890	16027	237.149

5.3.5 500 Job Instances (111–120)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
111	500	20	26699	3282.032	38312	3820.179
112	500	20	27163	3191.658	38723	3931.341
113	500	20	26985	3140.916	38251	3790.417
114	500	20	26934	3240.904	38820	4023.716
115	500	20	26816	3109.594	38341	3772.795
116	500	20	27076	3088.993	38667	3767.936
117	500	20	26757	3112.186	38297	3743.245
118	500	20	27179	3131.329	38507	3880.733
119	500	20	26545	3153.225	38277	3752.014
120	500	20	26901	3104.065	38319	3765.741

5.4 ACO Results

5.4.1 20 Job Instances (1–30)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
1	20	5	1297	84.926	1431	86.121
2	20	5	1365	84.041	1461	85.738
3	20	5	1161	86.292	1353	86.716
4	20	5	1329	83.751	1526	86.522
5	20	5	1269	84.117	1388	85.801
6	20	5	1224	83.398	1453	86.568
7	20	5	1253	98.678	1430	89.365
8	20	5	1227	85.591	1447	87.610
9	20	5	1272	84.433	1457	86.095
10	20	5	1146	83.667	1339	84.823
11	20	10	1668	116.481	1798	127.708
12	20	10	1750	120.431	1903	129.352
13	20	10	1558	116.816	1748	127.069
14	20	10	1418	114.629	1625	127.876
15	20	10	1502	115.115	1722	127.626
16	20	10	1453	115.647	1668	127.000
17	20	10	1542	123.030	1718	131.438
18	20	10	1620	118.569	1814	127.308
19	20	10	1664	123.296	1832	131.504
20	20	10	1656	123.661	1854	146.435
21	20	20	2399	208.068	2530	230.645
22	20	20	2127	195.506	2285	214.412
23	20	20	2401	191.673	2566	231.151
24	20	20	2262	198.337	2399	210.501
25	20	20	2397	199.022	2538	207.237
26	20	20	2325	204.382	2467	216.789
27	20	20	2362	195.013	2475	209.213
28	20	20	2249	189.245	2411	206.460
29	20	20	2306	193.830	2421	213.806
30	20	20	2277	190.420	2409	205.813

5.4.2 50 Job Instances (31–60)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
31	50	5	2730	456.507	3310	491.151
32	50	5	2882	465.186	3551	503.308
33	50	5	2625	466.329	3310	493.757
34	50	5	2783	461.610	3405	507.635
35	50	5	2876	467.099	3497	490.787
36	50	5	2842	461.491	3381	493.138
37	50	5	2802	476.917	3259	495.498
38	50	5	2688	467.745	3301	503.509
39	50	5	2575	467.898	3117	506.999
40	50	5	2786	461.859	3407	493.820
41	50	10	3133	672.868	3895	772.883
42	50	10	3042	672.165	3776	770.707
43	50	10	3029	674.378	3819	779.102
44	50	10	3158	685.742	3981	777.025
45	50	10	3154	664.200	3926	764.977
46	50	10	3127	673.591	3844	756.055
47	50	10	3311	672.361	3916	767.904
48	50	10	3140	674.454	3807	772.785
49	50	10	3017	683.283	3851	776.638
50	50	10	3258	682.641	3906	781.985
51	50	20	4054	1116.023	4846	1275.676
52	50	20	3909	1125.901	4635	1277.497
53	50	20	3913	1129.975	4669	1284.278
54	50	20	3956	1136.154	4724	1296.624
55	50	20	3906	1145.211	4475	1281.015
56	50	20	3888	1143.037	4566	1268.628
57	50	20	3968	1149.714	4600	1288.869
58	50	20	3966	1143.073	4756	1256.415
59	50	20	3918	1154.119	4576	1273.111
60	50	20	4015	1152.566	4793	1294.461

5.4.3 100 Job Instances (61–90)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
61	100	5	5535	1784.152	6762	1955.974
62	100	5	5288	1827.153	6568	1932.327
63	100	5	5219	1787.115	6431	1950.408
64	100	5	5023	1797.416	6172	1955.668
65	100	5	5266	1810.089	6442	1984.916
66	100	5	5139	1820.594	6323	1936.145
67	100	5	5272	1829.696	6578	1981.437
68	100	5	5129	1813.921	6331	1977.332
69	100	5	5489	1805.002	6693	1957.553
70	100	5	5334	1812.228	6586	1967.748
71	100	10	5947	2610.424	7558	3022.113
72	100	10	5402	2646.779	7339	3065.977
73	100	10	5740	2635.119	7434	3058.394
74	100	10	5904	2637.375	7682	3040.582
75	100	10	5599	2637.518	7343	3069.975
76	100	10	5409	2630.076	7121	3042.587
77	100	10	5745	2606.530	7363	3067.867
78	100	10	5712	2671.130	7316	3068.008
79	100	10	5998	2650.401	7525	3079.237
80	100	10	5923	2629.948	7562	3013.899
81	100	20	6625	4504.594	8389	5043.810
82	100	20	6592	4498.589	8329	5044.474
83	100	20	6634	4475.807	8288	5009.125
84	100	20	6621	4497.039	8333	5052.523
85	100	20	6685	4484.076	8251	5052.070
86	100	20	6722	4471.832	8360	5023.940
87	100	20	6580	4462.449	8542	5042.716
88	100	20	6800	4530.486	8555	5097.635
89	100	20	6662	4536.058	8303	5052.690
90	100	20	6767	4488.110	8472	5058.455

5.4.4 200 Job Instances (91–110)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
91	200	10	10976	10322.404	14480	12016.678
92	200	10	10639	10376.125	14433	11964.392
93	200	10	11072	10340.101	14338	12070.954
94	200	10	11057	10371.297	14429	11907.717
95	200	10	10613	10322.070	14426	11922.419
96	200	10	10449	10423.311	14172	12118.646
97	200	10	10960	10317.166	14604	11908.227
98	200	10	10871	10356.962	14562	11915.286
99	200	10	10558	10294.595	14316	12203.693
100	200	10	10794	11204.569	14412	12578.612
101	200	20	11620	18377.963	15768	21259.329
102	200	20	11749	18183.973	15759	20694.367
103	200	20	11839	18590.976	15993	20960.564
104	200	20	11816	18299.415	15932	20628.285
105	200	20	11628	18329.054	15855	20988.360
106	200	20	11717	18350.131	15911	21738.070
107	200	20	11776	19194.073	15900	21584.515
108	200	20	11855	18886.836	15875	20924.503
109	200	20	11659	18909.007	15997	22472.506
110	200	20	11893	18853.596	16027	21587.431

5.4.5 500 Job Instances (111–120)

Instance	Jobs	Machines	FSS Makespan	FSS Time (ms)	FSSB Makespan	FSSB Time (ms)
111	500	20	26699	116367.244	38312	131189.777
112	500	20	27163	115952.885	38723	137972.915
113	500	20	26985	119146.010	38251	140642.297
114	500	20	26934	119336.671	38820	130635.513
115	500	20	26816	114164.268	38341	130425.651
116	500	20	27076	113519.046	38667	130717.373
117	500	20	26757	114896.823	38297	135474.826
118	500	20	27179	112569.471	38507	131803.722
119	500	20	26545	114589.849	38277	130883.039
120	500	20	26901	116776.434	38319	128167.996

6 Comparison of NEH and ACO

To evaluate the effectiveness of the metaheuristic, the schedules generated by ACO were compared with the schedules generated by NEH.

Out of the 120 benchmark instances, ACO produced improved solutions in 13 instances. All improvements occurred within the 20-job problem class.

Table 1 lists all instances where ACO produced a better makespan than NEH.

Instance	Jobs	NEH Makespan	ACO Makespan
1	20	1319	1297
3	20	1165	1161
5	20	1305	1269
6	20	1269	1224
7	20	1256	1253
8	20	1188	1182
9	20	1239	1224
10	20	1206	1189
11	20	1244	1223
12	20	1213	1203
13	20	1248	1244
14	20	1231	1214
15	20	1239	1234

Table 1: Instances where ACO improved the NEH schedule

Although ACO was able to produce improved schedules in some instances, the improvements were relatively small.

Additionally, the computational cost of ACO was significantly higher. While NEH required less than one millisecond to generate a schedule, the ACO algorithm required approximately 80 to 100 milliseconds per 20 job instance.

7 Discussion

The results demonstrate that the NEH heuristic is highly effective for solving flow shop scheduling problems. Even for very large instances containing hundreds of jobs, NEH produces high-quality schedules with extremely small runtime.

The blocking flow shop problem consistently produced larger makespan values than the standard flow shop problem due to the additional constraints introduced by blocking.

Although the ACO metaheuristic improved several smaller instances, it did not outperform NEH for larger problems within the runtime limits used in this project. This suggests that additional iterations, parameter tuning, or hybrid metaheuristics may be required to produce improvements for large problem sizes.

8 Conclusion

In this project, the NEH heuristic was implemented to solve both the standard flow shop scheduling problem and the blocking flow shop scheduling problem across the 120 Taillard benchmark instances.

The results demonstrate that NEH is a powerful deterministic heuristic capable of producing high-quality schedules extremely quickly. The addition of an Ant Colony Optimization metaheuristic provided small improvements in some instances but required significantly more computational time.

Overall, NEH provides an excellent balance between solution quality and computational efficiency for flow shop scheduling problems.